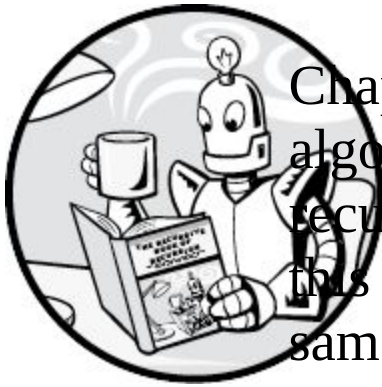


11

MAZE GENERATOR



Chapter 4 described a recursive algorithm that solves mazes, but another recursive algorithm generates mazes. In this chapter, we'll generate mazes in the same format as the maze-solver program in Chapter 4. So, whether you're a fan of solving mazes or creating them, you'll now have the power to apply programming to the task.

The algorithm works by visiting a starting space in the maze and then recursively visiting a neighboring space. The maze's hallways are “carved out” of the maze as the algorithm continues to visit neighbors. If the algorithm reaches a dead end that has no neighboring spaces, it backtracks to earlier spaces until it finds an unvisited neighbor and continues visiting from there. By the time the algorithm backtracks to the starting space, the entire maze has been generated.

The recursive backtracking algorithm we'll use here produces mazes that tend to have long hallways (the maze spaces that connect branching intersections) and are fairly simple to solve. However, this algorithm is easier to implement than many other maze-generation algorithms, such as Kruskal's algorithm or Wilson's algorithm, so it serves as a good introduction to the topic.